

The exploration cost of boundary and population effects

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Abstract

Can a growing deployment support causal inference while the operational cost of experimentation vanishes? The answer depends on the effect being estimated. We study known assignment policies that increasingly favor a preferred action and measure exploration by the cumulative operational value forgone through alternative actions. In one model with a one-dimensional score and mean outcomes that are Lipschitz near the decision boundary, a shrinking common temperature permits pointwise asymptotically valid inference for the boundary effect while exploration loss tends to zero. The same model contains outcome laws with identical boundary effects but different population average effects that become indistinguishable under vanishing loss. Uniform mean-square consistency for the population effect requires divergent worst-case loss. In a larger model without extrapolation restrictions, an information–exploration inequality applies to all estimators and adaptive designs, with a lower-bound constant involving the operational gap functional $(E\sqrt{g})^2$. For sufficiently small fixed target error, one deterministic common temperature requires loss growing at least as $n/\log^2 n$, while uniform mixing has bounded loss. Conversational AI motivates the setting; the formal results concern a single strategy assignment per independent unit, with the text-generation kernel held fixed.

Keywords: adaptive experiments, limited overlap, regression discontinuity, minimax lower bounds, generative AI, off-policy evaluation

MSC 2020: 62D20, 62K05, 62C20, 62G20

1 Introduction

A deployment can generate increasingly many observations while providing very different information about different causal effects. Under a shrinking common temperature, alternative actions become concentrated near the decision boundary. Those comparisons can

support local inference at a decreasing operational cost. Learning an average effect over all contexts requires comparisons elsewhere as well. This paper quantifies that difference using one outcome model and one exploration-loss criterion.

Conversational artificial intelligence provides a motivating example. In randomized trials, GPT-4 shifted debate opponents’ positions more often than human debaters did [1], and conversations with frontier models moved policy attitudes with effects that persisted for weeks [2]. Establishing that a technology persuades is the first question, not the last. Regulation and replication concern narrower objects: whether presenting evidence rather than asking a question, at this point in this conversation, changes what a person believes. Systems that persuade at scale choose such strategies adaptively, and their logs are what an analyst inherits. One of the strongest levers identified by Hackenburg et al. [2], a reward model that selects among candidate responses, is itself an adaptive selection policy.

This paper asks a precise version of the question those logs raise. Suppose the deployer’s strategy policy is known, as it is when the deployer records action probabilities, and suppose it sharpens as the deployer becomes confident, for instance by lowering a softmax temperature. Which causal effects can the logs estimate, and what must be paid to estimate the others?

The answer depends on the target. We distinguish two:

- the *boundary effect*, the effect among contexts where the policy is indifferent between strategies; and
- the *population average effect* over all contexts the deployment serves.

These answer different questions. The first is relevant to marginal changes of a policy that is already deployed; the second to the average outcome change from assigning strategy 1 rather than strategy 0 across the full context population, holding the generator fixed. Estimating one well does not substitute for estimating the other.

That local and population effects differ, and that limited overlap makes average effects hard to estimate, are familiar facts. What this paper adds is a comparison of the two targets inside one outcome model and on one cost scale: the boundary effect admits valid inference while the cumulative cost of exploration tends to zero, whereas no estimator learns the population effect uniformly without divergent cost. Estimability becomes a statement about what an experiment must pay for.

We price learning in the deployer’s own currency. Each time the logged strategy differs from the one the deployer’s objective prefers, the deployer gives up an operational value gap $g(h)$. The cumulative sum is the *exploration loss* R_n . This quantity is defined by the deployer’s objective, not by the human outcome under study, so it can be computed at the design stage without knowing the effects being estimated.

Figure 1 previews the construction behind Theorem 3. Perturbing only the unpreferred action away from the boundary changes the population effect but leaves the boundary effect unchanged. A sharpening policy produces very few observations of precisely that action in the perturbed region.

Contributions.

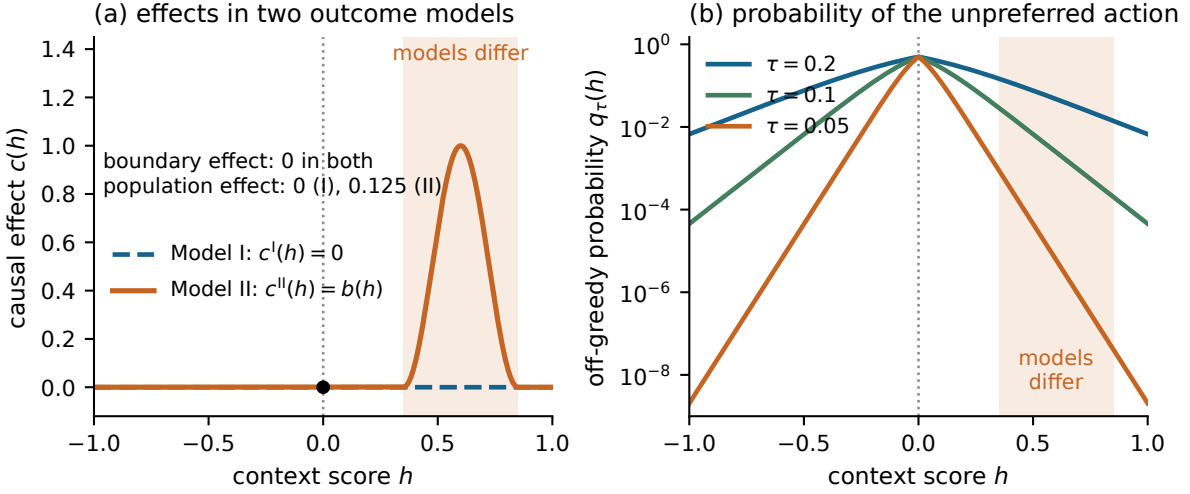


Figure 1: The separation construction, with $H \sim U(-1, 1)$, $g(h) = |h|$, $u_1 = 0.35$, $u_2 = 0.85$, and $t = 1$. Left: the two effect functions $c^0 = 0$ and $c^1 = b$ (superscripts index outcome laws) have the same boundary effect, zero, but population effects 0 and 0.125. The greedy-arm mean is unchanged. Right: the probability of an off-greedy action under three illustrative temperatures. Shading marks the region in which the outcome laws differ. These are exact functions from the construction, not simulation estimates or fitted effects.

1. *Separation in one model* (Theorem 3). In a one-dimensional score model with Lipschitz mean outcomes near the decision boundary, logs generated by a single common temperature $\tau_n = n^{-\zeta}$, $\zeta \in (1/2, 1)$, give asymptotically valid Wald intervals for the boundary effect at every member of the model, while $R_n \rightarrow 0$ (Theorem 1). In the same model, whenever $R_n \rightarrow 0$, two members that coincide near the boundary but differ in the population effect have total variation distance tending to zero, so no estimator is consistent for the population effect at both; and uniform consistency for the population effect requires $\sup R_n \rightarrow \infty$. Figure 2 shows the separation in computed risks for a simple design.
2. *An information-exploration inequality* (Theorem 2). In the model of all bounded mean functions, for every estimator and every adaptive or non-adaptive design, the product of worst-case mean squared error and worst-case exploration loss plus a constant is at least $\sigma^2(\sum_k f_k \sqrt{g_k})^2$. Uniform precision δ therefore requires worst-case exploration loss at least of order $(\mathbb{E}\sqrt{g})^2/\delta^2$ as the target error tends to zero; a design exploring with probability proportional to $g^{-1/2}$ attains this order up to constants (Proposition 1).
3. *What this means for temperature*. For any sufficiently small fixed precision δ , a single deterministic common temperature requires exploration loss at least of order $n/\log^2(n\delta^2)$ (Theorem 4), against a loss of order δ^{-2} , bounded in n , for the designs below. Context-dependent temperatures implement the gap-based design (Proposi-

tion 2). Random mixtures of finite temperatures can also avoid the rate (Remark 1). When the value gap is misspecified, a Kantorovich bound says when gap-based allocation is worth its risk relative to uniform mixing (Lemma 1).

Scope. All results are for one decision per unit with independent contexts and a known assignment policy. Theorem 1 and Theorem 3 use a one-dimensional score, and their boundary guarantees are pointwise in the outcome law. The impossibility results hold in models without extrapolation structure: where data near the boundary identify the population effect, as under a constant effect, the separation disappears. Multi-turn conversations, estimated policies, and effects of realized text features are outside the formal results.

Relation to existing work. *Boundary estimands from algorithmic assignment.* Narita and Yata [3] show that weights of overlap type, built from an approximate propensity score of a known algorithm, converge to a boundary estimand and require undersmoothing; this is the closest antecedent for Theorem 1. There the smoothing is an analyst-chosen ball in covariate space; here it is a temperature chosen by the system in score space, and we compute the exploration loss jointly with the inferential guarantee. Eckles et al. [4] use known assignment noise near a cutoff for design-based inference, with a fixed noise scale and no cost accounting. Overlap weights and their target population are due to Li, Morgan and Zaslavsky [5]; they note that the weights peak where the propensity score is one half, but do not study a sharpening limit. Humphreys [6] shows that the fixed-effects estimand weights stratum effects by the assignment variance $p_j(1 - p_j)$, the same weighting, and can differ from the average effect; under a sharpening policy this weighting concentrates on the decision boundary.

Exploration and estimation. Simchi-Levi and Wang [7] show a regret–estimation trade-off in multi-armed bandits without contexts; Wan, Kveton and Song [8] optimize exploration for policy evaluation under a value constraint; Douglas, Persson and Provost [9] derive optimal logging policies for inverse-propensity estimation and report that softmax logging performs worse than top- k logging in simulation; Koppel et al. [10] extend inverse-gap-weighting contextual bandits to balance regret against the quality of average-treatment-effect estimates (we have read only the abstract of that workshop paper). The allocation $q \propto g^{-1/2}$ has the form of optimum stratified allocation with costs [11], and Ravichandran et al. [12] derive the analogous cost-weighted rule, sample sizes proportional to $S_j/\sqrt{C_j}$, for factorial experiments under a budget. We do not claim the allocation as new; what we add is that the corresponding constant bounds every estimator and adaptive design when loss is paid only for off-greedy actions. Li and Imai [13] compare evaluating an individualized treatment rule by randomizing units to the rule with evaluation from a standard experiment, without a cost for assignments against the rule; our exploration loss makes that cost explicit. Cesa-Bianchi et al. [14] show that Boltzmann exploration with a common schedule can fail for regret and that arm-dependent schedules repair it; our Theorem 4 and Proposition 2 are estimation analogues.

Limited overlap and off-policy evaluation. Lower bounds for off-policy value estimation

[15–17] and limited-overlap asymptotics [18–20] supply the information-theoretic ingredients. Khan and Tamer [18] observe that the average treatment effect loses identification when any region of the covariate support is removed, and derive rates for trimmed inverse weighting when propensities approach zero or one in the covariate tails; their propensities do not change with n , and there is no cost of exploration. What we add is the coupling of the information bound to the exploration loss, and the comparison of boundary and population targets on that common scale. Ring and Schomaker [21] give a diagnostic for stochastic positivity problems and use it to choose alternative estimands when the data do not support the original one; our results are a formal counterpart for known, sharpening assignment policies, stating which estimand remains learnable and at what exploration cost. Khan, Saveski and Ugander [22] bound effects without overlap under smoothness, which is the kind of extrapolation structure our impossibility excludes.

Generative treatments. Athey, Imbens and Ji [23] identify effects of content features from stochastic algorithms using logged unexposed candidates and probability ratios. Their estimand averages over co-occurring pairs with trimming and does not conflict with our bounds, which concern the population effect of a strategy under outcomes observed only for the logged action. Fresh and Shin [24] give a taxonomy of conversational estimands; we work at the level of strategy assignment. Imai and Nakamura [25] and Nakamura and Imai [26] study effects of realized text features under deterministic decoding; we instead hold the generation kernel fixed.

2 Setup

Generative treatments. Units $i = 1, \dots, n$ arrive with contexts $H_i \sim F$, independently. The system chooses a strategy $A_i \in \{0, 1\}$ and produces a message $M_i \sim G(\cdot \mid A_i, H_i)$ from a fixed generation kernel G , which includes token-level sampling settings. The outcome is Y_i . The treatment is assignment of the strategy with G held fixed: potential outcomes $Y_i(a)$ are defined under the kernel intervention that sets $A_i = a$ and draws the message from G [27, 28]. A label attached to a realized message after the fact is a different object, and changing G , for example by changing token temperature, changes the treatment. We assume:

- (C1) *consistency*: $Y_i = Y_i(A_i)$, where the randomness of the message drawn from G is part of $Y_i(a)$;
- (C2) *independent units*: $(H_i, Y_i(0), Y_i(1))$ are independent and identically distributed across i , with no interference;
- (C3) *known sequential assignment*: given H_i and the past data $\mathcal{F}_{i-1}$, A_i is drawn with known probabilities, independently of $(Y_i(0), Y_i(1))$ and of all future units.

Under (C1)–(C3), $\mathbb{E}[Y_i \mid A_i = a, H_i = h, \mathcal{F}_{i-1}] = \mathbb{E}[Y(a) \mid H = h]$, and the law of Y_i given $(A_i = a, H_i = h, \mathcal{F}_{i-1})$ is that of $Y(a)$ given $H = h$. All statements below about conditional outcome laws are therefore statements about potential outcomes.

Deployer. The deployer has a score gap $\Delta(h) = s_1(h) - s_0(h)$ from its own objective, a greedy strategy $a^*(h) = \mathbf{1}\{\Delta(h) > 0\}$, and an operational value gap $g(h) \geq 0$, the value its objective loses by choosing $\bar{a}(h) = 1 - a^*(h)$. We write $g = \phi(|\Delta|)$ when the gap is a function of the score.

Running example. A conversational system chooses between presenting evidence ($A = 1$) and asking a question ($A = 0$). A reward model scores both strategies given the conversation so far, H ; $\Delta(H)$ is the difference of the two scores and the system logs the probability $e(H)$ with which it chose evidence. The operational gap $g(H)$ is the loss in the deployer’s own objective, for example predicted engagement, from using the lower-scoring strategy. The outcome Y is the person’s attitude after the conversation, measured separately and not used by the policy. The generator that writes the message under each strategy is held fixed. The theory below does not use anything specific to language models beyond this structure.

Designs and notation for probabilities. We use two probabilities and keep them distinct:

$$e(h) = \mathbb{P}(A = 1 \mid H = h), \quad q(h) = \mathbb{P}(A \neq a^*(h) \mid H = h),$$

the probability of strategy 1 and the probability of the off-greedy strategy. A design specifies $Q_i = \mathbb{P}(A_i = \bar{a}(H_i) \mid H_i, \mathcal{F}_{i-1})$; it is non-adaptive if $Q_i = q(H_i)$. A *common temperature* design uses one deterministic $\tau = \tau_n$ and

$$e_\tau(h) = \text{sig}(\Delta(h)/\tau), \quad q_\tau(h) = \text{sig}(-|\Delta(h)|/\tau),$$

where $\text{sig}(x) = 1/(1 + \exp(-x))$.

Exploration loss. We measure expected cumulative operational loss, rather than loss per unit:

$$R_n = \mathbb{E} \left[\sum_{i=1}^n g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\} \right]. \quad (1)$$

For a non-adaptive design, $R_n = n\mathbb{E}[g(H)q(H)]$. The context law F , the score Δ , and the operational gap g are fixed when comparing outcome laws; under an adaptive design, R_n can nevertheless depend on the outcome law through past observations.

Estimands. Let $\mu_a(h) = \mathbb{E}[Y(a) \mid H = h]$ and $c = \mu_1 - \mu_0$. The population average effect is $\theta = \mathbb{E}[c(H)]$. With a one-dimensional score $\Delta(h) = h$, the boundary effect is $\beta_0 = c(0)$. For a given temperature, the overlap-weighted effect $\beta_{\text{ov},\tau} = \mathbb{E}[\omega_\tau(H)c(H)]/\mathbb{E}[\omega_\tau(H)]$ with $\omega_\tau = e_\tau(1 - e_\tau)$ [5] is a third, temperature-indexed, target; we use it only as an intermediate object.

3 The result in brief

Take a one-dimensional score, $\Delta(h) = h$, and outcome laws whose mean functions are bounded and Lipschitz near the decision boundary (Assumption 1). Let the logs come from one common temperature $\tau_n = n^{-\zeta}$ with $\zeta \in (1/2, 1)$.

- (i) The boundary effect β_0 has an asymptotically valid Wald interval, and the cumulative exploration loss R_n tends to zero (Theorem 1).
- (ii) In the same model, outcome laws with equal boundary effects but different population effects cannot be distinguished whenever $R_n \rightarrow 0$, and uniformly consistent estimation of θ requires divergent worst-case loss (Theorem 3).
- (iii) Without extrapolation structure, precision δ for θ , uniformly over the model, requires worst-case loss of order at least $(\mathbb{E}\sqrt{g})^2/\delta^2$, whichever design and estimator are used, and this does not decrease with n (Theorem 2).

Sections 4–6 state these results precisely, and Section 7 draws the consequences for temperature as a design lever.

Figure 2 illustrates the separation for the design used in Section 8, with risks computed by numerical integration: the exact risk of the Horvitz–Thompson estimator of θ , which is unbiased, and a first-order approximation to the root mean squared error of the boundary estimator, $\{(\beta_{\text{ov},\tau} - \beta_0)^2 + V_1/n\}^{1/2}$, where V_1 is its linearization variance from the proof of Theorem 1. Across 16 Monte Carlo checks with $n\tau \geq 3$ (4,000 replications each), the Monte Carlo root mean squared error is between 0.994 and 1.023 times the approximation; at $n\tau = 0.33$ the ratio is 0.53, so the approximation overstates the error there. Under one common temperature, the population effect reaches root mean squared error 0.1 only where the exploration loss is large: on the plotted grid the smallest such loss is about 102. The boundary effect reaches the same precision with loss 0.58 at the edge of the grid, and with less loss at larger n and sharper policies.

4 Boundary effects at vanishing exploration loss

Assumption 1 (Boundary model). $\Delta(h) = h$, $a^*(h) = \mathbf{1}\{h > 0\}$, and

- (D1) H has a density $f \leq \bar{f}$, continuous at 0, with $f(0) > 0$;
- (D2) $|\mu_a| \leq B$, and μ_a is L -Lipschitz on $[-u_0, u_0]$;
- (D3) $\sigma_a^2(h) = \text{Var}(Y \mid A = a, H = h)$ is continuous at 0 with $\sigma_1^2(0) + \sigma_0^2(0) > 0$, and $\mathbb{E}[Y^4 \mid A, H] \leq M$;
- (D4) $\mathbb{P}(A = 1 \mid H = h) = \text{sig}(h/\tau_n)$ with $\tau_n \rightarrow 0$, known;
- (D5) $g = \phi(|h|) \leq \bar{G}$ with $\phi(u) \leq \bar{\kappa}u$ on $[0, u_0]$; in the strong form $\phi'(0) = \kappa > 0$.

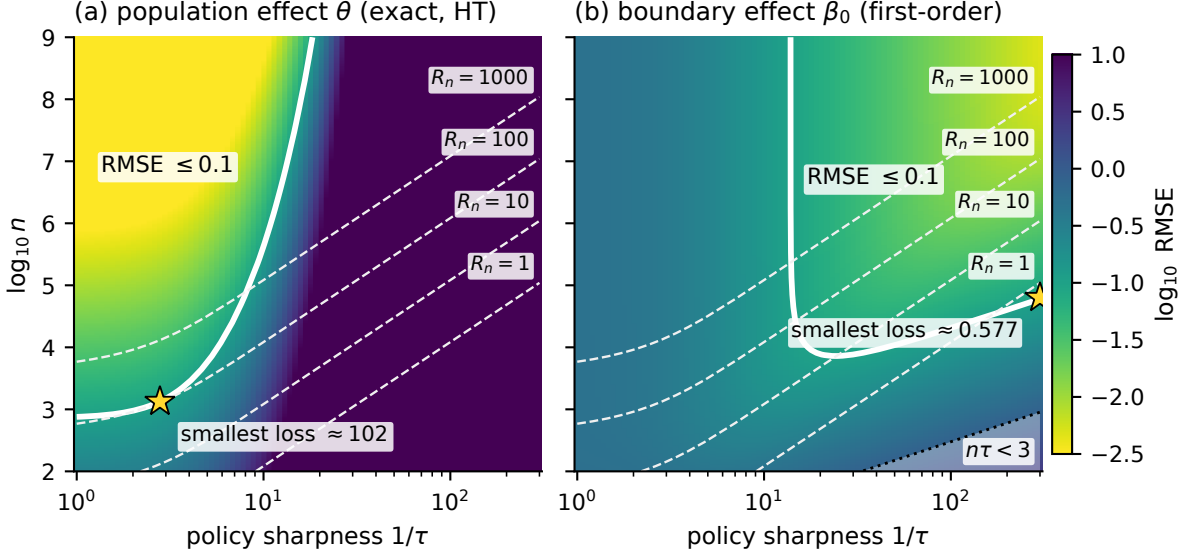


Figure 2: Root mean squared error (colour, $\log_{10}$ scale) under one common temperature τ for the design of Section 8: $H \sim U(-1, 1)$, $\Delta(h) = h$, $g(h) = |h|$, $c(h) = 1 + |h|$, unit noise. (a) Exact root mean squared error of the Horvitz–Thompson estimator of the population effect θ . (b) First-order approximation, including the exact bias, to the root mean squared error of the estimator of Theorem 1 for the boundary effect β_0 ; the shaded region below the dotted line has $n\tau < 3$, where Monte Carlo checks show the approximation overstates the error. White curves enclose the regions with root mean squared error at most 0.1. Dashed curves are levels of the cumulative exploration loss $R_n = n\mathbb{E}[g(H)q_\tau(H)]$. Stars mark the smallest loss that attains the precision on the plotted grid ($1 \leq 1/\tau \leq 300$, $10^2 \leq n \leq 10^9$). Panel (a) concerns one estimator under one outcome law; the lower bounds for all estimators are Theorems 2 and 3.

With $e_i = e_{\tau_n}(H_i)$, define the overlap weights and arm means by

$$\begin{aligned} w_{1i} &= A_i(1 - e_i), & w_{0i} &= (1 - A_i)e_i, \\ \hat{D}_a &= \frac{1}{n} \sum_{i=1}^n w_{ai}, & \hat{m}_a &= \frac{\sum_{i=1}^n w_{ai} Y_i}{\sum_{i=1}^n w_{ai}}, & \hat{\beta} &= \hat{m}_1 - \hat{m}_0. \end{aligned} \quad (2)$$

For inference use

$$\begin{aligned} \hat{\psi}_i &= \frac{w_{1i}(Y_i - \hat{m}_1)}{\hat{D}_1} - \frac{w_{0i}(Y_i - \hat{m}_0)}{\hat{D}_0}, \\ \hat{s}^2 &= \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i^2, & \widehat{\text{SE}}(\hat{\beta}) &= \frac{\hat{s}}{\sqrt{n}}. \end{aligned} \quad (3)$$

If either denominator is zero, set $\hat{m}_a$, $\hat{\beta}$, $\hat{s}^2$ and the undefined influence values to zero, and report the interval $\mathbb{R}$. This event has probability tending to zero under the theorem's conditions; it does not change the asymptotic conclusions.

Theorem 1 (Boundary inference). *Under Assumption 1, if $n\tau_n \rightarrow \infty$:*

- (i) $(\hat{\beta} - \beta_{\text{ov}, \tau_n}) / (s_n / \sqrt{n}) \Rightarrow N(0, 1)$ with $s_n^2 = \{\sigma_1^2(0) + \sigma_0^2(0)\} / \{2\tau_n f(0)\} (1 + o(1))$;
- (ii) $|\beta_{\text{ov}, \tau_n} - \beta_0| \leq C\tau_n$ for large n ;
- (iii) $\hat{s}^2 / s_n^2 \rightarrow 1$ in probability;
- (iv) if also $n\tau_n^3 \rightarrow 0$, the Wald interval $\hat{\beta} \pm z_{1-\eta/2} \hat{s} / \sqrt{n}$ has asymptotic coverage $1 - \eta$ for β_0 ;
- (v) $R_n = n\mathbb{E}[g(H)q_{\tau_n}(H)] \leq n\tau_n^2 \bar{\kappa} \bar{f} \pi^2 / 6 + n\bar{G} \exp(-u_0 / \tau_n)$, and in the strong form $R_n = n\tau_n^2 \kappa f(0) \pi^2 / 6 (1 + o(1))$.

In particular, for $\tau_n = n^{-\zeta}$ with $\zeta \in (1/2, 1)$, the interval in (iv) is asymptotically valid and $R_n \rightarrow 0$.

The three rate conditions are $n\tau_n \rightarrow \infty$ (the boundary sample grows, $\zeta < 1$), $n\tau_n^3 \rightarrow 0$ (bias is negligible relative to the standard error, $\zeta > 1/3$), and $n\tau_n^2 \rightarrow 0$ (the loss vanishes, $\zeta > 1/2$). The error $|\hat{\beta} - \beta_0| = O_p((n\tau_n)^{-1/2})$ is achievable; we do not claim it is optimal, and it is a statement in probability rather than in mean square. Differentiability of c at zero does not improve the bias order in general: a bounded effect equal to $c(h) = 1 + |h|^{3/2}$ near zero gives bias proportional to $\tau^{3/2}$. If c is $C^{1,1}$ and f is Lipschitz near zero, the bias is $O(\tau^2)$ and condition (iv) relaxes to $n\tau_n^5 \rightarrow 0$. Proofs are in Appendix A.

Vanishing loss does not mean that exploration disappears. The expected number of off-greedy actions is $n\mathbb{E}[q_{\tau_n}(H)] = 2 \log 2 f(0) n\tau_n (1 + o(1))$, which grows because $n\tau_n \rightarrow \infty$; the loss nevertheless vanishes, $R_n = O(n\tau_n^2)$, with asymptotic equivalence in the strong form of (D5), because these actions occur ever closer to the boundary, where the value gap is small.

Assumption (D5) is what makes boundary exploration cheap: the deployer's objective is nearly indifferent near the boundary, so choosing the unpreferred strategy there costs little. It does not describe a fixed cost per exploratory action, such as a review or payment incurred whenever the policy deviates. If every off-greedy action costs at least $c_0 > 0$, then $R_n \geq c_0 n \mathbb{E}[q_{\tau_n}(H)] = 2 \log 2 c_0 f(0) n \tau_n (1 + o(1))$, which diverges under the condition $n \tau_n \rightarrow \infty$ of Theorem 1. Under such costs, boundary inference also requires divergent cumulative loss, and the separation below does not hold in this form.

5 Population effects require paid exploration

Fix $B > 0$ and $\sigma > 0$. Let $\mathcal{M}(B, \sigma)$ be the model in which $Y \mid A = a, H = h$ is $N(\mu_a(h), \sigma^2)$ and μ_0, μ_1 are *any* measurable functions bounded by B . The model contains no structure that carries effects learned near the boundary to other contexts.

Theorem 2 (Information–exploration inequality). *Let $S_1, \dots, S_K$ be disjoint sets with $f_k = F(S_k) > 0$, each contained in $\{a^* = 0\}$ or in $\{a^* = 1\}$, and $g_k = \inf_{S_k} g$. For every design, adaptive or not, and every estimator $\hat{\theta}$,*

$$\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \cdot \left(\sup_{\mathcal{M}(B, \sigma)} R_n + \frac{\sigma^2 \pi^2}{B^2} \sum_k g_k \right) \geq \sigma^2 \left(\sum_k f_k \sqrt{g_k} \right)^2. \quad (4)$$

Consequently, if $F(g \geq \gamma) > 0$ for some $\gamma > 0$, uniform consistency requires $\sup R_n \rightarrow \infty$; and if $\sup \mathbb{E}(\hat{\theta} - \theta)^2 \leq \delta_n^2 \rightarrow 0$, then $\liminf_n \delta_n^2 \sup R_n \geq \sigma^2 (\mathbb{E} \sqrt{g(H)})^2$.

Neither side of (4) involves n : deployment scale does not substitute for exploration when the target is the population effect. If $\sum_k g_k > 0$, at $R_n = 0$ the bound is the finite floor $B^2 (\sum f_k \sqrt{g_k})^2 / (\pi^2 \sum g_k)$, consistent with the fact that a constant estimator has bounded error.

Proposition 1 (Achievability). *Suppose $0 < \mathbb{E} \sqrt{g} < \infty$, $\delta_n > 0$, and $n \delta_n^2 \rightarrow \infty$. Write $C = \sigma^2 + B^2$. For $n \delta_n^2 > 4C$, define*

$$c_n = \frac{\mathbb{E} \sqrt{g}}{n \delta_n^2 / C - 4}, \quad q_n(h) = \begin{cases} \min\{1/2, c_n / \sqrt{g(h)}\}, & g(h) > 0, \\ 1/2, & g(h) = 0. \end{cases} \quad (5)$$

Let $e_n(h)$ be the corresponding probability of strategy 1. Then

$$\hat{\theta}_{\text{HT}} = \frac{1}{n} \sum_{i=1}^n \left\{ \frac{A_i Y_i}{e_n(H_i)} - \frac{(1 - A_i) Y_i}{1 - e_n(H_i)} \right\} \quad (6)$$

has $\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta}_{\text{HT}} - \theta)^2 \leq \delta_n^2$, and

$$R_n \leq \frac{C n (\mathbb{E} \sqrt{g})^2}{n \delta_n^2 - 4C} = C (\mathbb{E} \sqrt{g})^2 \delta_n^{-2} (1 + o(1)).$$

If also $\mathbb{E}[g] < \infty$, uniform mixing with $q \equiv C/(n\delta_n^2 - 2C)$, for $n\delta_n^2 \geq 4C$, has the same uniform risk guarantee and

$$R_n = \frac{Cn\mathbb{E}[g]}{n\delta_n^2 - 2C} = C\mathbb{E}[g]\delta_n^{-2}(1 + o(1)).$$

The order δ^{-2} and the functional $(\mathbb{E}\sqrt{g})^2$ therefore appear in both the lower bound and the attainable loss. These upper bounds are constructive guarantees; they do not assert exact finite-sample optimality of the allocation. The constants differ: the lower bound has σ^2 , and the Horvitz–Thompson upper bound retains B^2 . We do not establish the optimal constant. Proofs are in Appendix B.

6 The separation

Theorem 2 is proved in $\mathcal{M}(B, \sigma)$, while Theorem 1 assumes Lipschitz means near the boundary. A lower bound over a large model does not transfer to a smaller one automatically. We therefore state both sides in one model.

Let $\mathcal{M}_D$ be the set of outcome laws satisfying (D1), (D2), Gaussian noise with variance σ^2 , and (D5) with ϕ not necessarily monotone. Under Gaussian noise, (D3) holds for every member of $\mathcal{M}_D$.

Assumption 2. There are $u_0 < u_1 < u_2$ with $F([u_1, u_2]) > 0$ and $\gamma := \text{ess inf}_{h \in [u_1, u_2]} g(h) > 0$, and $B, L > 0$.

Theorem 3 (Separation). *Under Assumption 2, let $b(h) = \sin^2(\pi(h - u_1)/(u_2 - u_1))\mathbf{1}\{u_1 \leq h \leq u_2\}$, $t_{\max} = \min\{B, L(u_2 - u_1)/\pi\}$, and for $|t| \leq t_{\max}$ let $\mu^t \in \mathcal{M}_D$ have $\mu_1^t \equiv 0$ and $\mu_0^t = -tb$. All μ^t coincide on $(-\infty, u_1]$, have boundary effect $\beta_0 = 0$, and have population effect $\theta(\mu^t) = t\mathbb{E}[b(H)]$.*

(S1) *Under the common-temperature design $\tau_n = n^{-\zeta}$, $\zeta \in (1/2, 1)$, $R_n \rightarrow 0$ and, for every $\mu \in \mathcal{M}_D$, the Wald interval of Theorem 1 has asymptotic coverage $1 - \eta$ for $\beta_0(\mu)$.*

(S2) *For every design and every t , $\text{TV}(P_{\mu^0}^n, P_{\mu^t}^n) \leq R_n(\mu^0)/\gamma$. Hence if $R_n(\mu^0) \rightarrow 0$, for every $t_1 \in (0, t_{\max}]$ and every estimator, $\max_{t \in \{0, t_1\}} P_{\mu^t}(|\hat{\theta}_n - \theta(\mu^t)| \geq t_1\mathbb{E}[b]/2) \geq (1 - R_n(\mu^0)/\gamma)/2 \rightarrow 1/2$.*

(S3) *For every design and every estimator,*

$$\sup_{\mathcal{M}_D} \mathbb{E}(\hat{\theta} - \theta)^2 \geq \frac{(\mathbb{E}[b])^2}{\sup_{\mathcal{M}_D} R_n/(\sigma^2\gamma) + \pi^2/t_{\max}^2}.$$

Under the design of (S1), μ^0 and μ^{t_1} are both members of $\mathcal{M}_D$ with the same boundary effect, and the boundary interval is valid at each. By (S2), no estimator of the population effect is consistent at both. By (S3), consistency uniformly over $\mathcal{M}_D$ requires unbounded exploration loss. We do not claim that the constant of Theorem 2 holds in $\mathcal{M}_D$, nor any uniform guarantee on the boundary side. If a submodel lets boundary data identify θ , the perturbation family is excluded and (S2)–(S3) do not apply. Proofs are in Appendix C.

7 Temperature as an exploration device

Theorem 4 (One common temperature). *Consider common-temperature designs with one deterministic τ_n per n . Assume $g = \phi(|\Delta|)$, $\mathbb{E}[g] < \infty$, and $|\Delta(H)|$ has a density at least $f_{\min} > 0$ on $[0, u_0]$; $\phi(u) \geq \kappa u$ on $[0, u_0]$; $D_{\max} = \text{ess sup } |\Delta(H)| < \infty$; and for every $\eta' \in (0, D_{\max})$, $\nu_{\eta'} = \max_j \mathbb{P}(|\Delta(H)| \geq D_{\max} - \eta', a^*(H) = j) > 0$. Fix $\eta' \in (0, D_{\max})$. If an estimator has $\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \leq \delta_n^2$ with $\delta_n^2 \leq \nu_{\eta'}^2 B^2 / (2\pi^2)$ and $2n\delta_n^2 > \nu_{\eta'} \sigma^2$, then*

$$R_n \geq \kappa f_{\min} c_0 n \min \left\{ u_0^2, \frac{(D_{\max} - \eta')^2}{\log^2 \{2n\delta_n^2 / (\nu_{\eta'} \sigma^2)\}} \right\}, \quad c_0 = \int_0^1 v \text{sig}(-v) dv.$$

This is a lower bound, and δ_n may be fixed. For any fixed δ with $\delta^2 \leq \nu_{\eta'}^2 B^2 / (2\pi^2)$, the common-temperature loss is at least of order $n / \log^2 n$, whereas Proposition 1 attains the same uniform precision with loss bounded in n . The precision condition cannot be dropped: for $\delta \geq 2B$ the estimator that always outputs zero meets the target with no exploration. Relative to the attainable loss, the lower bound is larger by a factor of order at least $n\delta_n^2 / \log^2(n\delta_n^2)$, which diverges whenever $n\delta_n^2 \rightarrow \infty$. With finitely many score values $0 < \Delta_{\min} < \Delta_{\max}$ and fixed, sufficiently small δ , the common-temperature loss is at least of order $n^{1-\Delta_{\min}/\Delta_{\max}}$. The reason is not that a common temperature explores in the wrong place: the optimal design also explores most near the boundary, where the operational gap is small. It is that the exploration probability of a common temperature decays exponentially in $|\Delta|$, so observations far from the boundary become too rare unless the temperature is raised everywhere, which is expensive near the boundary.

Remark 1 (Mixtures of temperatures). Theorem 4 does not extend to random mixtures of finite temperatures. With $|\Delta| = |H|$, $H \sim U(-1, 1)$, $g = |H|$, mixing temperature 1 with weight $w_n = \mathbb{E}[1/\text{sig}(-|H|)] / (n\delta^2)$ and temperature $1/n$ otherwise keeps the design variance $\mathbb{E}[1/q]/n$ at most δ^2 , for fixed δ , with loss bounded in n (46.36 at $\delta = 0.1$ for $n = 10^4, 10^5, 10^6$). The lesson concerns a single common temperature, not temperature as such.

Proposition 2 (Context temperatures). *For $\Delta(h) \neq 0$ and $q(h) \in (0, 1/2)$, the context temperature $\tau(h) = |\Delta(h)| / \log\{(1 - q(h))/q(h)\}$ gives off-greedy probability $q(h)$. The value $q(h) = 1/2$ requires $\tau(h) = \infty$, and on $\{\Delta = 0\}$ every temperature gives $q = 1/2$. Suppose $g = \phi(|\Delta|)$ with $\phi(0) = 0$, so that $g = 0$ on $\{\Delta = 0\}$. Then the design q^* of Proposition 1 is implementable with temperatures in $(0, \infty]$: $q^* = 1/2$ wherever $g = 0$, and for each h with $\Delta(h) \neq 0$ and $g(h) > 0$, $q^*(h) < 1/2$ for large n and $\tau^*(h) \log(1/c_n) \rightarrow |\Delta(h)|$.*

The limit is pointwise for fixed h with $g(h) > 0$. It does not say that the implementing temperature is small near the boundary at finite n : where the cap binds, for example $0 < |h| \leq 4c_n^2$ when $g(h) = |h|$, the implementing temperature is $\tau = \infty$.

Lemma 1 (Misspecified gaps). *Assume $F(g > 0) = 1$, $0 < \mathbb{E}\sqrt{g} < \infty$, and $0 < a \leq \hat{g}/g \leq b < \infty$. Compare the designs $q = c/\sqrt{g}$ and $q = c'/\sqrt{g}$, with c, c' chosen so that both have the same sparse-exploration criterion $\sigma^2 \mathbb{E}[1/q]/n$, and suppose both probabilities*

are at most $1/2$ almost surely. Let $\rho = b/a$. Then the loss ratio is at most $K(\rho) = (1 + \sqrt{\rho})^2/(4\sqrt{\rho})$. The bound is attained when the measure $w \propto \sqrt{g} dF$ has a set of mass exactly one half, for example when it is non-atomic.

A common rescaling of $\hat{g}$ changes nothing; only relative misspecification across contexts matters. Gap-based allocation beats uniform mixing for every admissible $\hat{g}$ if $\mathbb{E}[g]/(\mathbb{E}\sqrt{g})^2 > K(\rho)$; the condition is also necessary when the bound is attained, but not in general. The criterion $\sigma^2\mathbb{E}[1/q]/n$ is not the full estimator MSE. For capped designs, including gaps with positive probability of being zero, the ratio bound holds as the inverse-exploration moment tends to infinity. Suppose $0 < a \leq \hat{g}/g \leq b < \infty$ on $\{g > 0\}$, $\hat{g} = 0$ on $\{g = 0\}$, $0 < \mathbb{E}\sqrt{g} < \infty$, and let $q_c = \min\{1/2, c/\sqrt{\hat{g}}\}$ with $q_c = 1/2$ where $\hat{g} = 0$. By dominated convergence, $c\mathbb{E}[1/q_c] = \mathbb{E}\max\{2c, \sqrt{\hat{g}}\} \rightarrow \mathbb{E}\sqrt{\hat{g}}$ and $\mathbb{E}[gq_c]/c \rightarrow \mathbb{E}[g/\sqrt{\hat{g}}]$ (with $g/\sqrt{\hat{g}} = 0$ where $g = 0$) as $c \rightarrow 0$. Calibrating c to a common inverse-exploration moment $T = \mathbb{E}[1/q_c] \rightarrow \infty$, the loss ratio converges to $\mathbb{E}[\sqrt{\hat{g}}]\mathbb{E}[g/\sqrt{\hat{g}}]/(\mathbb{E}\sqrt{g})^2$, the quantity bounded in the lemma. Proofs are in Appendix D.

8 Simulations

The Monte Carlo simulations use $H \sim U(-1, 1)$, $\Delta(h) = h$, $g(h) = |h|$, $\mu_0(h) = h$, $c(h) = 1 + |h|$ (population effect 1.5, boundary effect 1), and $N(0, 1)$ noise.

Exploration designs. For the allocation comparisons we match the sparse-exploration criterion $V_{\text{sp}}(q) = \sigma^2\mathbb{E}[1/q]/n$; it is an allocation criterion, not an identity for finite-sample MSE. At $n = 10^5$ and $V_{\text{sp}} = \delta^2 = 0.01$, the closed-form exploration loss is 50.0 for uniform mixing, 44.4 for $q \propto g^{-1/2}$, and 988.2 for a common temperature ($\tau = 0.110$), a ratio of 19.8 to uniform mixing. Table 1 runs the augmented inverse-propensity estimator with a linear working model on common random numbers across designs. The context-temperature implementation of $q \propto g^{-1/2}$ produces identical rows and is not listed separately. The designs are matched on V_{sp} , not on realized RMSE. The working model is correctly specified for this data-generating process, so the table illustrates the designs rather than verifying the uniform Horvitz–Thompson guarantee of Proposition 1. At fixed δ , the expected number of off-greedy actions stays bounded for uniform mixing and the gap-based design, so increasing n alone does not establish nominal Wald coverage for those designs. This bounded-count statement does not apply to the common-temperature design. Figure 3 shows the allocation costs across deployment sizes at $V_{\text{sp}} = 0.1^2$; Figure 5 in Appendix E repeats it at $V_{\text{sp}} = 0.03^2$ with the same ordering. In a two-context example the bound (4) evaluates to 51.9 against the asymptotic constant 52.4, and the common-temperature loss has local growth exponent 0.808 against the predicted 0.8.

Boundary inference. Table 2 reports the boundary estimator of Theorem 1. Exploration loss matches the constant in Theorem 1(v). With $\zeta = 0.6$ coverage is close to nominal; at $n = 10^6$, 5,000 replications give 0.9458 (MCSE 0.0032), with the estimated

δ	design	RMSE	coverage	MCSE	explored	loss R_n
0.10	uniform mixing	0.098	0.950	0.011	100	49.9
0.10	$q \propto g^{-1/2}$ (= context temp.)	0.103	0.932	0.013	134	44.5
0.10	common temperature	0.097	0.965	0.009	7607	988.5
0.05	uniform mixing	0.052	0.955	0.010	400	200.3
0.05	$q \propto g^{-1/2}$ (= context temp.)	0.053	0.940	0.012	532	178.0
0.05	common temperature	0.050	0.948	0.011	9191	1440.1

Table 1: Exploration designs at $n = 10^5$, 400 replications with common random numbers. Coverage is of the population effect by Wald intervals; MCSE uses the observed coverage. “Explored” is the mean number of off-greedy actions.

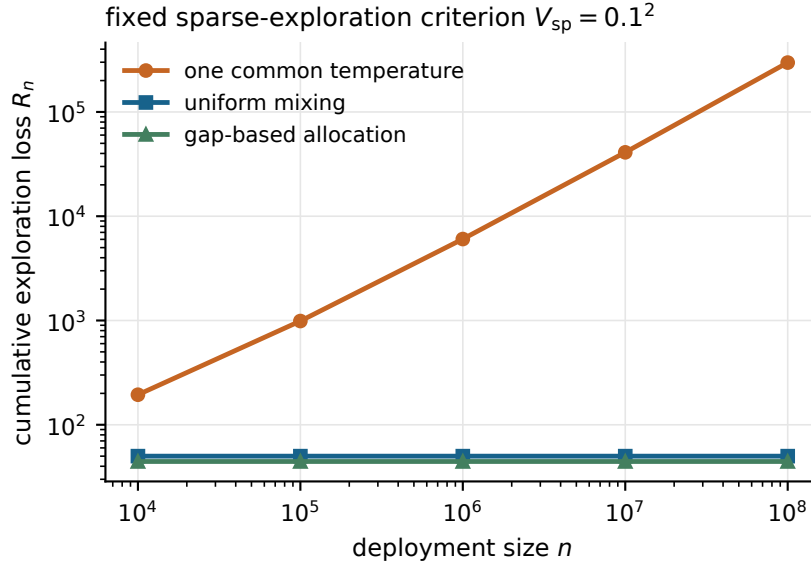


Figure 3: Cumulative exploration loss as deployment size increases, from closed-form calculations with $H \sim U(-1, 1)$, $g(h) = |h|$, $\sigma^2 = 1$ and the sparse-exploration criterion fixed at $V_{\text{sp}} = 0.1^2$. Designs are matched on this criterion, not on realized finite-sample error. A context-dependent temperature implements the gap-based probabilities and gives the same curve.

standard error at 0.993 of the Monte Carlo standard deviation. With a very small boundary sample the interval under-covers: 0.887 at $n\tau_n \approx 10$. The guarantee of Theorem 1 is asymptotic.

Temperature sweep. Figure 4 fixes $n = 100,000$ and sharpens a common temperature, with 1,000 replications per temperature. Wald coverage of the population effect falls from 0.949 at $1/\tau = 1$ to 0.085 (Horvitz–Thompson) and 0.031 (augmented estimator with a fitted linear working model) at $1/\tau = 120$. Over the same range, coverage of the boundary effect rises from 0.000 to 0.943, and the exploration loss falls from 17,056 to 5.7. Boundary coverage improves over this range because $n\tau$ stays at least 833. This is not a monotone property of sharpening: at fixed n , lowering the temperature further drives $n\tau$ toward zero, and the boundary interval then fails as well, as Theorem 1 requires $n\tau_n \rightarrow \infty$ and Table 2 shows at $n\tau \approx 10$.

Error diagnostics for the same runs are in Figure 6 in Appendix E. The Monte Carlo error of the Horvitz–Thompson estimator understates its exact standard deviation once the policy is sharp: at $1/\tau = 120$ the Monte Carlo root mean squared error of the Horvitz–Thompson estimator is 0.36, while its exact standard deviation is of order 10^{22} . Its variance is carried by rare, very large terms that the replications do not observe. For the Horvitz–Thompson estimator, the exact standard deviation rather than the Monte Carlo points describes the error. The oracle augmented estimator, which uses the true regression, is shown only as an exact reference; its variance is not the variance of the fitted augmented estimator whose Monte Carlo results are plotted. For the boundary estimator the line is the first-order approximation described in Section 6, and the Monte Carlo points agree with it.

ζ	n	$n\tau_n$	reps	RMSE	coverage of β_0	MCSE	loss R_n (Thm. 1)
0.60	10,000	39.8	1,000	0.215	0.949	0.007	1.3×10^{-1} (1.3×10^{-1})
0.60	100,000	100.0	500	0.141	0.944	0.010	8.2×10^{-2} (8.2×10^{-2})
0.60	1,000,000	251.2	200	0.082	0.970	0.012	5.1×10^{-2} (5.2×10^{-2})
0.75	10,000	10.0	1,000	0.464	0.887	0.010	8.1×10^{-3} (8.2×10^{-3})
0.75	100,000	17.8	500	0.339	0.924	0.012	2.6×10^{-3} (2.6×10^{-3})
0.75	1,000,000	31.6	200	0.247	0.930	0.018	8.2×10^{-4} (8.2×10^{-4})
0.60	1,000,000	251.2	5,000	0.090	0.946	0.003	–

Table 2: Boundary inference under common temperature $\tau_n = n^{-\zeta}$. Coverage is of $\beta_0 = c(0)$. Loss is the Monte Carlo mean with the Theorem 1(v) constant in parentheses. The last row is a separate 5,000-replication run.

9 Discussion

The results separate two uses of adaptive logs. If the question is how outcomes respond where the deployed policy is indifferent, under the boundary assumptions, a growing deployment permits increasingly precise local inference while cumulative exploration loss

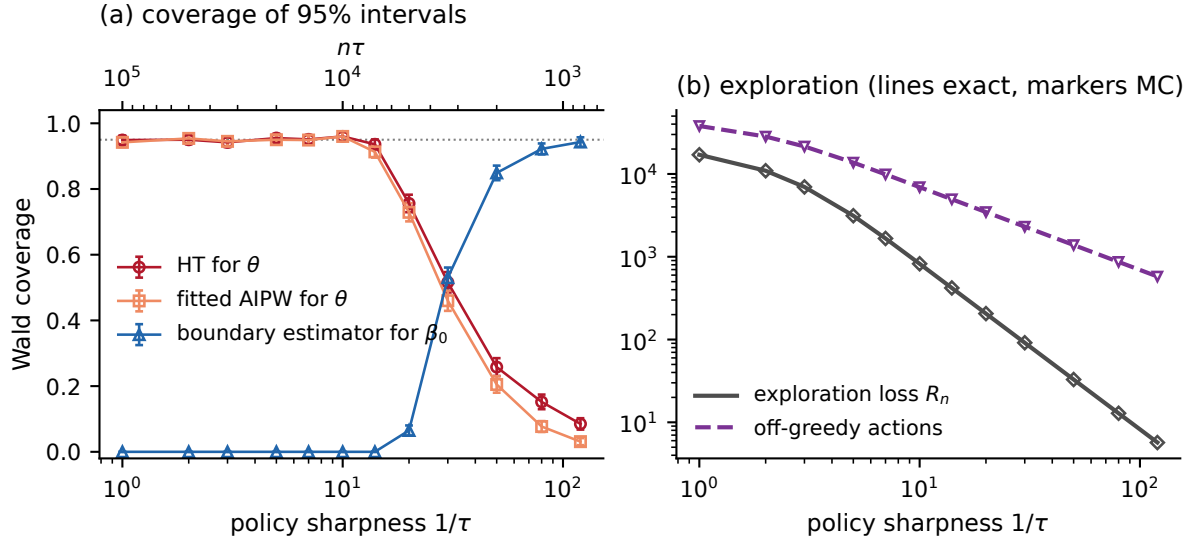


Figure 4: Common temperature at fixed $n = 100,000$, 1,000 replications per temperature, same design as Figure 2. (a) Wald coverage with 95% Monte Carlo intervals for the population effect (Horvitz–Thompson and augmented estimator with a fitted linear working model) and for the boundary effect; the top axis shows $n\tau$. Boundary coverage is low while $n\tau^3$ is large, as Theorem 1(iv) requires. (b) Cumulative exploration loss R_n and expected number of off-greedy actions; lines are exact expectations, markers Monte Carlo.

vanishes. The number of exploratory actions still grows; their average operational cost falls. If the question is the population average effect of a strategy, uniformly consistent estimation over the model requires divergent worst-case exploration loss. Increasing deployment size does not remove that requirement. Reporting an effect estimated from adaptive logs therefore requires saying which of the two quantities it is.

For design, the relevant choice is not whether to use temperature but how. With one temperature shared by all contexts, the exploration probability decays exponentially in the policy’s confidence, so reaching confident contexts forces a temperature that is costly near the boundary; uniform mixing, context-dependent temperatures, or temperature mixtures with a vanishing weight on a moderate temperature avoid this. Among designs that avoid it, allocating by the value gap saves further loss when gaps are heterogeneous and reasonably well specified (Lemma 1); in our example the saving is about ten percent.

Limitations. The policy is known and assignments are logged; the setting is a single decision per unit with independent contexts; the boundary results are one-dimensional and pointwise; the population impossibility excludes models in which boundary data identify the population effect; and the treatment is strategy assignment under a fixed generator, not the realized text. Extending the separation to multi-turn conversations requires defining the reference policy for later turns and the induced distribution of histories, which we leave open. Estimated policies, multidimensional scores and effects of realized text features are also outside the formal results.

Data availability statement

No new participant data were collected or analysed. The results are theoretical, deterministic numerical illustrations, or simulations. The simulation code, the result files (with arguments, random seeds and code commits), plain-text versions of the proofs, and the scripts that generate every number, table and figure in the paper are available in the repository

<https://github.com/kunhailP/exploration-cost-boundary-population>

(tagged submission version [TO CONFIRM: tag]).

Funding information

The author states no funding involved.

Author contribution

The author confirms the sole responsibility for the conception of the study, presented results and manuscript preparation.

Conflict of interest

The author states no conflict of interest.

Ethical approval

The conducted research is not related to either human or animals use.

Use of AI-assisted tools

[TO CONFIRM against the publisher’s policy] Generative AI tools were used to assist in drafting text, checking proofs, and writing simulation code. The author reviewed all content and takes full responsibility for it.

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A Proof of Theorem 1

Write $e = e_\tau(H) = \text{sig}(H/\tau) = \mathbb{P}(A = 1 \mid H)$ and $\omega = e(1 - e) = K(H/\tau)$ with $K(u) = \exp(u)/\{1 + \exp(u)\}^2$, and $m_\tau = \mathbb{E}\omega$. We use

$$\begin{aligned} \int K &= 1, & \int |u|K(u) du &= 2 \log 2, & \int K(u)\text{sig}(u) du &= \frac{1}{2}, \\ K(u) &\leq \exp(-|u|), & \int_0^\infty v \text{sig}(-v) dv &= \frac{\pi^2}{12}. \end{aligned}$$

Step 1 (kernel integrals). For ϱ bounded and continuous at 0, $j \geq 1$ and $r, s \geq 0$, substituting $h = \tau u$ and using dominated convergence with dominating function $\|\varrho\|_\infty \bar{f} K(u)$,

$$\mathbb{E}[K(H/\tau)^j (1 - e)^r e^s \varrho(H)] = \tau f(0) \varrho(0) \int K^j \text{sig}(-u)^r \text{sig}(u)^s du + o(\tau).$$

In particular $m_\tau = \tau f(0)(1 + o(1))$.

Step 2 (bias). $\beta_{\text{ov},\tau} - \beta_0 = \mathbb{E}[\omega\{c(H) - c(0)\}]/m_\tau$. On $|H| \leq u_0$, $|c(h) - c(0)| \leq 2L|h|$ and $\mathbb{E}[\omega|H|] \leq \bar{f}\tau^2 \int |u|K = 2 \log 2 \bar{f}\tau^2$. On $|H| > u_0$, $\omega \leq \exp(-u_0/\tau)$ and $|c(h) - c(0)| \leq 4B$. Hence $|\beta_{\text{ov},\tau} - \beta_0| \leq (4 \log 2 L \bar{f}\tau^2 + 4B \exp(-u_0/\tau))/m_\tau = (4 \log 2 L \bar{f}/f(0))\tau(1 + o(1))$.

Step 3 (linearization). Let $\bar{\mu}_a = \mathbb{E}[\omega\mu_a]/m_\tau$, so $\bar{\mu}_1 - \bar{\mu}_0 = \beta_{\text{ov},\tau}$. Since $\mathbb{E}[w_1 | H] = \omega$ and $\mathbb{E}[w_1 Y | H] = \omega\mu_1$, $\mathbb{E}[w_1(Y - \bar{\mu}_1)] = 0$, and likewise for arm 0. Define $T_a = n^{-1} \sum_i w_{ai}(Y_i - \bar{\mu}_a)/m_\tau$ and $\xi_i = \{w_{1i}(Y_i - \bar{\mu}_1) - w_{0i}(Y_i - \bar{\mu}_0)\}/m_\tau$. Then $\hat{m}_a - \bar{\mu}_a = (m_\tau/\hat{D}_a)T_a$ and $\mathbb{E}\xi = 0$. Because $w_1 w_0 = 0$ and $\mathbb{E}[A(1-e)^2 | H] = \omega(1-e)$,

$$s_n^2 := \text{Var}\xi = m_\tau^{-2} \mathbb{E}[\omega\{(1-e)(\sigma_1^2 + (\mu_1 - \bar{\mu}_1)^2) + e(\sigma_0^2 + (\mu_0 - \bar{\mu}_0)^2)\}].$$

By Step 1, $\bar{\mu}_a \rightarrow \mu_a(0)$; the terms in $(\mu_a - \bar{\mu}_a)^2$ are $o(\tau)$ by dominated convergence; and the variance terms equal $\tau f(0)\sigma_a^2(0)/2 + o(\tau)$. Hence $s_n^2 = \{\sigma_1^2(0) + \sigma_0^2(0)\}/\{2\tau f(0)\}(1 + o(1))$.

Step 4 (central limit theorem). Since $0 \leq w_a \leq 1$, $w_a^4 \leq w_a$, and $\mathbb{E}[(Y - \bar{\mu}_a)^4 | A, H] \leq C(M, B)$, we get $\mathbb{E}[w_a^4(Y - \bar{\mu}_a)^4] \leq C m_\tau$ and $\mathbb{E}\xi^4 \leq C' m_\tau^{-3}$. The Lyapunov ratio $n^{-1}\mathbb{E}\xi^4/s_n^4 = O((n\tau)^{-1}) \rightarrow 0$, so $n^{-1/2} \sum \xi_i/s_n \Rightarrow N(0, 1)$. Also $\text{Var}(\hat{D}_a)/m_\tau^2 \leq 1/(nm_\tau) \rightarrow 0$, so $\hat{D}_a/m_\tau \rightarrow 1$ in probability and $m_\tau/\hat{D}_a - 1 = O_p((n\tau)^{-1/2})$. Now

$$\hat{\beta} - \beta_{\text{ov},\tau} = (T_1 - T_0) + (m_\tau/\hat{D}_1 - 1)T_1 - (m_\tau/\hat{D}_0 - 1)T_0,$$

with $T_1 - T_0 = n^{-1} \sum \xi_i$ and $T_a = O_p((n\tau)^{-1/2})$. The remainder is $O_p((n\tau)^{-1}) = o_p(s_n/\sqrt{n})$, and Slutsky's lemma gives (i).

Step 5 (variance estimator). Write $\xi_{ai} = w_{ai}(Y_i - \bar{\mu}_a)/m_\tau$. First, $n^{-1} \sum \xi_i^2/s_n^2$ has mean one and variance at most $\mathbb{E}\xi^4/(ns_n^4) = O((n\tau)^{-1})$. Second,

$$\hat{\psi}_{1i} - \xi_{1i} = w_{1i}(Y_i - \bar{\mu}_1)(1/\hat{D}_1 - 1/m_\tau) + w_{1i}(\bar{\mu}_1 - \hat{m}_1)/\hat{D}_1.$$

The mean square of the first term is $(n^{-1} \sum \xi_{1i}^2)(m_\tau/\hat{D}_1 - 1)^2 = O_p(s_n^2)O_p((n\tau)^{-1})$. For the second, $w_1^2 \leq w_1$ gives mean square at most $(\hat{m}_1 - \bar{\mu}_1)^2/\hat{D}_1$; by Step 4 and $\hat{D}_1 = \tau f(0)(1 + o_p(1))$ this is $O_p(1/(n\tau^2))$, which is $O_p((n\tau)^{-1})$ relative to $s_n^2 \asymp \tau^{-1}$. Arm 0 is identical. By Minkowski's and the Cauchy-Schwarz inequalities, $|n^{-1} \sum \hat{\psi}_i^2 - n^{-1} \sum \xi_i^2| = o_p(s_n^2)$, which gives (iii).

Step 6 (centering at β_0). $(\beta_{\text{ov},\tau} - \beta_0)/(s_n/\sqrt{n}) = O(\tau)O((n\tau)^{1/2}) = O((n\tau^3)^{1/2})$. With (i), (iii) and Slutsky's lemma this gives (iv).

Step 7 (exploration loss). The off-greedy probability is $q_\tau(h) = \text{sig}(-|h|/\tau)$, which equals $e_\tau(h)$ for $h < 0$ and $1 - e_\tau(h)$ for $h > 0$. On $|h| \leq u_0$, $n\mathbb{E}[\phi(|H|)\text{sig}(-|H|/\tau)] \leq n\bar{\kappa}\bar{f} \cdot 2\tau^2 \int_0^\infty v \text{sig}(-v)dv = n\bar{\kappa}\bar{f}\tau^2\pi^2/6$, and the rest is at most $n\bar{G}\exp(-u_0/\tau)$. In the strong form, on $v \leq u_0/\tau$, $\phi(\tau v)/\tau \rightarrow \kappa v$ and $f(\tau v) \rightarrow f(0)$ with dominating function $\bar{\kappa}v f \text{sig}(-v)$, which gives the asymptotic equality. For $\tau_n = n^{-\zeta}$: $n\tau_n \rightarrow \infty$ iff $\zeta < 1$; $n\tau_n^3 \rightarrow 0$ iff $\zeta > 1/3$; $n\tau_n^2 \rightarrow 0$ iff $\zeta > 1/2$. $\square$

Smoother effects. If $|c(h) - c(0) - c'(0)h| \leq M_2 h^2$ and f is L_f -Lipschitz near 0, then $\mathbb{E}[\omega H] = \tau^2 \int u K(u) \{f(\tau u) - f(0)\} du = O(\tau^3)$ up to an exponentially small tail, and the quadratic remainder contributes at most $M_2 \bar{f} \tau^3 \int u^2 K/m_\tau$; the bias is $O(\tau^2)$. Differentiability of c alone is not enough: a bounded effect equal to $c(h) = 1 + |h|^{3/2}$ near zero gives bias $\tau^{3/2} \int |u|^{3/2} K(u) du (1 + o(1))$.

B Proofs for Section 5

Proof of Theorem 2. *Submodel.* For $\vartheta \in [-B, B]^K$ set $\mu_{a^*(h)}^\vartheta(h) = 0$, and on S_k set $\mu_{\bar{a}(h)}^\vartheta(h) = s_k \vartheta_k$ with $s_k = +1$ if $\bar{a} = 1$ on S_k and $s_k = -1$ otherwise; elsewhere $\mu_{\bar{a}(h)}^\vartheta(h) = 0$. Then $\mu^\vartheta \in \mathcal{M}(B, \sigma)$ and $\theta(\mu^\vartheta) = \sum_k f_k \vartheta_k$.

Information. The sample density is $\prod_i dF(H_i) \kappa_i(A_i \mid H_i, \mathcal{F}_{i-1}) \varphi_\sigma(Y_i - \mu_{A_i}^\vartheta(H_i))$, where the assignment kernels κ_i are known functions of observed data. The score for ϑ_k is $\sum_i \mathbf{1}\{H_i \in S_k, A_i = \bar{a}(H_i)\} s_k (Y_i - s_k \vartheta_k) / \sigma^2$. Its summands are martingale differences, so the Fisher information matrix is diagonal with $I_k(\vartheta) = \mathbb{E}_\vartheta N_k / \sigma^2$, where $N_k = \sum_i \mathbf{1}\{H_i \in S_k, A_i = \bar{a}(H_i)\}$.

van Trees. Take the product prior with marginal density $B^{-1} \cos^2(\pi x / (2B))$ on $[-B, B]$, whose Fisher information is $J = \pi^2 / B^2$. With $\bar{N}_k = \int \mathbb{E}_\vartheta N_k \lambda(d\vartheta)$, the multivariate van Trees inequality [29], maximized over the direction vector, gives

$$\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \geq \sum_k \frac{f_k^2}{\bar{N}_k / \sigma^2 + J} = \sum_k \frac{f_k \sigma^2}{a_k}, \quad a_k = \frac{\bar{N}_k + \sigma^2 J}{f_k}.$$

Cauchy-Schwarz and loss. Writing $\sigma \sum_k f_k \sqrt{g_k} = \sum_k \sqrt{f_k g_k a_k} \cdot \sigma \sqrt{f_k / a_k}$,

$$\sigma^2 \left(\sum_k f_k \sqrt{g_k} \right)^2 \leq \left(\sum_k f_k g_k a_k \right) \left(\sum_k f_k \sigma^2 / a_k \right).$$

Here $\sum_k f_k g_k a_k = \sum_k g_k \bar{N}_k + \sigma^2 J \sum_k g_k$, and since $g \geq g_k$ on S_k , $\sum_k g_k N_k \leq \sum_i g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\}$, so $\sum_k g_k \bar{N}_k \leq \sup R_n$. This proves (4).

Consequences. With $K = 1$ and $S_1 = \{g \geq \gamma\} \cap \{a^* = j\}$ of positive mass, (4) gives $\sup R_n \geq g_1 \{\sigma^2 f_1^2 / \sup \mathbb{E}(\hat{\theta} - \theta)^2 - \sigma^2 \pi^2 / B^2\} \rightarrow \infty$. For the constant, fix a partition into level sets of g intersected with the two greedy sides; (4) gives $\liminf \delta_n^2 \sup R_n \geq \sigma^2 (\sum f_k \sqrt{g_k})^2$, and lower sums $\sum F(S) \inf_S \sqrt{g}$ increase to $\mathbb{E} \sqrt{g}$ as the level mesh shrinks. $\square$

Proof of Proposition 1. The Horvitz-Thompson estimator is unbiased with variance at most $n^{-1}(\sigma^2 + B^2) \mathbb{E}[1/e + 1/(1-e)] = n^{-1}(\sigma^2 + B^2) \mathbb{E}[1/q + 1/(1-q)] \leq n^{-1}(\sigma^2 + B^2)(\mathbb{E}[1/q] + 2)$, since $\{e, 1-e\} = \{q, 1-q\}$ and $q \leq 1/2$. Pointwise $1/q = \max\{2, \sqrt{g}/c_n\} \leq \sqrt{g}/c_n + 2$, and the choice of c_n gives $\mathbb{E}[1/q] + 2 \leq n \delta_n^2 / (\sigma^2 + B^2)$. The loss is $n \mathbb{E}[gq] \leq n c_n \mathbb{E} \sqrt{g}$. For uniform mixing with $C = \sigma^2 + B^2$ and $q = C / (n \delta_n^2 - 2C) \leq 1/2$, the same variance bound gives $C(1/q + 2)/n = \delta_n^2$, and the loss is $n q \mathbb{E}[g] = C \mathbb{E}[g] n / (n \delta_n^2 - 2C)$. Without the correction, $q = C / (n \delta_n^2)$ gives risk $C / \{n q (1-q)\} > \delta_n^2$ at $\mu_0 = \mu_1 \equiv B$, where the bound is attained. $\square$

C Proof of Theorem 3

The function b satisfies $0 \leq b \leq 1$, is $\pi/(u_2 - u_1)$ -Lipschitz, vanishes on $(-\infty, u_1]$, and has $\mathbb{E}b(H) > 0$ because H has a density and $F([u_1, u_2]) > 0$. Hence $|tb| \leq B$ and tb

is L -Lipschitz for $|t| \leq t_{\max}$, so $\mu^t \in \mathcal{M}_D$; $c^t = tb$, $\beta_0(\mu^t) = 0$, and $\theta(\mu^t) = t\mathbb{E}b(H)$. Every perturbed observation is off-greedy, since $a^* = 1$ on $(0, \infty)$, with $H \in [u_1, u_2]$. Let $N = \sum_i \mathbf{1}\{A_i = 0, H_i \in [u_1, u_2]\}$.

(S1) is Theorem 1, because every member of $\mathcal{M}_D$ satisfies Assumption 1.

(S2). The sample densities under μ^0 and μ^t coincide on $E = \{N = 0\}$, so $P_0(E) = P_t(E)$, and for any event A , $|P_0(A) - P_t(A)| = |P_0(A \cap E^c) - P_t(A \cap E^c)| \leq \max\{P_0(E^c), P_t(E^c)\} = P_0(E^c)$. Thus $\text{TV} \leq P_0(N \geq 1) \leq \mathbb{E}_0 N$. Because $g \geq \gamma$ F -almost surely on $[u_1, u_2]$ and $H_i \sim F$, $\gamma N \leq \sum_i g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\}$ almost surely, so $\mathbb{E}_0 N \leq R_n(\mu^0)/\gamma$. Le Cam's two-point argument with separation $t_1 \mathbb{E}b$ gives the displayed bound.

(S3). The score of $t \mapsto \mu^t$ is $\sum_i \mathbf{1}\{A_i = 0\}(-b(H_i))(Y_i + tb(H_i))/\sigma^2$, with information $I(t) = \mathbb{E}_t \sum_i \mathbf{1}\{A_i = 0\}b(H_i)^2/\sigma^2 \leq \mathbb{E}_t N/\sigma^2 \leq R_n(\mu^t)/(\gamma\sigma^2)$. With the prior $t_{\max}^{-1} \cos^2(\pi t/(2t_{\max}))$ on $[-t_{\max}, t_{\max}]$, whose information is $\pi^2/t_{\max}^2$, the van Trees inequality for $\psi(t) = t\mathbb{E}b$ gives the bound. $\square$

D Proofs for Section 7

Proof of Theorem 4. Take the single cell $S = \{|\Delta| \geq D_{\max} - \eta', a^* = j\}$, with j attaining $\nu = \nu_{\eta'}$. There $q_\tau \leq \exp(-(D_{\max} - \eta')/\tau)$, so $\bar{N} \leq n\nu \exp(-(D_{\max} - \eta')/\tau)$, and the van Trees step of Theorem 2 with $K = 1$ gives $\delta_n^2 \geq \nu^2 \sigma^2 / (n\nu \exp(-(D_{\max} - \eta')/\tau) + \sigma^2 \pi^2 / B^2)$. Once $\delta_n^2 \leq \nu^2 B^2 / (2\pi^2)$, this forces $n\nu \exp(-(D_{\max} - \eta')/\tau) \geq \nu^2 \sigma^2 / (2\delta_n^2)$, that is, $\tau_n \geq (D_{\max} - \eta') / \log\{2n\delta_n^2 / (\nu\sigma^2)\}$, where the logarithm is positive because $2n\delta_n^2 > \nu\sigma^2$. Nothing here requires $\delta_n \rightarrow 0$. On the other hand

$$R_n = n\mathbb{E}[\phi(|\Delta|)\text{sig}(-|\Delta|/\tau)] \geq n\kappa f_{\min} \tau^2 \int_0^{u_0/\tau} v \text{sig}(-v) dv \geq n\kappa f_{\min} c_0 \min\{\tau, u_0\}^2.$$

Combining the two bounds proves the claim. With finitely many score values, the same argument gives $\tau \geq \Delta_{\max} / \log(Cn\delta^2)$ for δ small enough. The $\Delta_{\min}$ cell, with mass f , then contributes at least

$$nf\phi(\Delta_{\min}) \exp(-\Delta_{\min}/\tau)/2. \quad \square$$

Remark 1. Write $u = |H|$, $r(u) = \text{sig}(-u)$ for the off-greedy probability at temperature 1, and $q_n = w_n r + (1 - w_n) \text{sig}(-nu)$ for the mixture. Since $q_n \geq w_n r$, $\mathbb{E}[1/q_n]/n \leq \delta^2$ for $w_n = \mathbb{E}[1/r]/(n\delta^2)$, and $n\mathbb{E}[uq_n] \leq \mathbb{E}[1/r]\mathbb{E}[ur]/\delta^2 + O(1/n)$. With $w_n = \mathbb{E}[1/r]/\{n\delta_n^2/(\sigma^2 + B^2) - 2\}$ the Horvitz–Thompson risk is at most δ_n^2 uniformly over $\mathcal{M}(B, \sigma)$.

Proof of Proposition 2. Solving $\text{sig}(-|\Delta|/\tau) = q$ gives the formula. If $g(h) = 0$, then $q^*(h) = 1/2$, which is implementable by $\tau = \infty$ (and by any τ when $\Delta(h) = 0$). If $\Delta(h) \neq 0$ and $g(h) > 0$, then $q^*(h) = c_n/\sqrt{g(h)} < 1/2$ once $c_n < \sqrt{g(h)}/2$, and $\log\{(1 - q^*)/q^*\} = \log(1/c_n) + \frac{1}{2} \log g(h) + \log(1 - q^*)$ with $c_n \rightarrow 0$, which gives the limit. $\square$

Proof of Lemma 1. With the cap slack, let $q = c/\sqrt{\hat{g}}$ and $\mathbb{E}[1/q] = T$. The loss is $n\mathbb{E}[\sqrt{\hat{g}}]\mathbb{E}[g/\sqrt{\hat{g}}]/T$, against $n(\mathbb{E}\sqrt{g})^2/T$ for $\hat{g} = g$. With $s = (\hat{g}/g)^{1/2} \in [\sqrt{a}, \sqrt{b}]$ and the probability measure $w \propto \sqrt{g} dF$, the ratio is $\mathbb{E}_w[s]\mathbb{E}_w[1/s]$, which Kantorovich's inequality bounds by $(\sqrt{a} + \sqrt{b})^2/(4\sqrt{ab}) = K(b/a)$. If w has a set of mass one half, setting $s = \sqrt{a}$ there and $s = \sqrt{b}$ elsewhere attains the bound. With atoms the bound need not be attained: for $F = (0.99, 0.01)$, $g = (1, 100)$ and $\rho = 36$, uniform mixing costs 1.675 times the optimum, below $K(36) = 2.042$, yet every admissible $\hat{g}$ costs at most 1.347 times the optimum. $\square$

E Additional figures

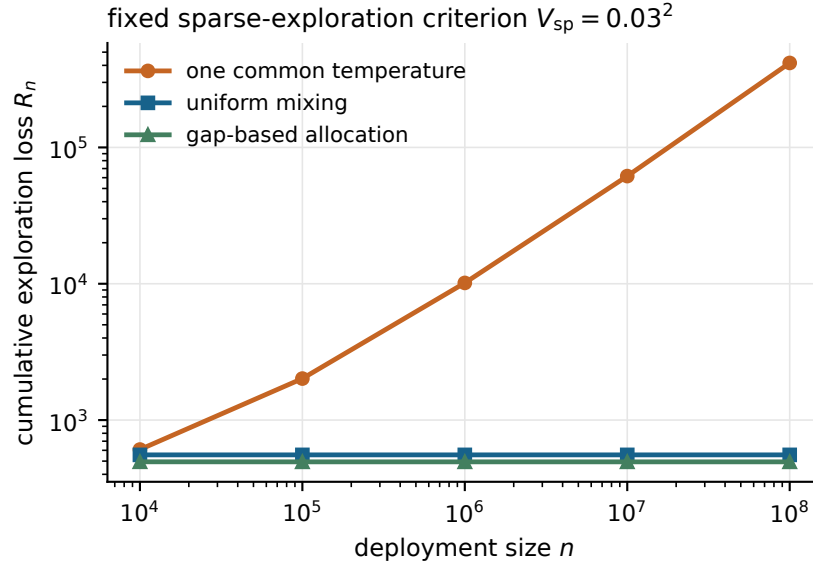


Figure 5: As Figure 3, with the sparse-exploration criterion fixed at $V_{\text{sp}} = 0.03^2$.

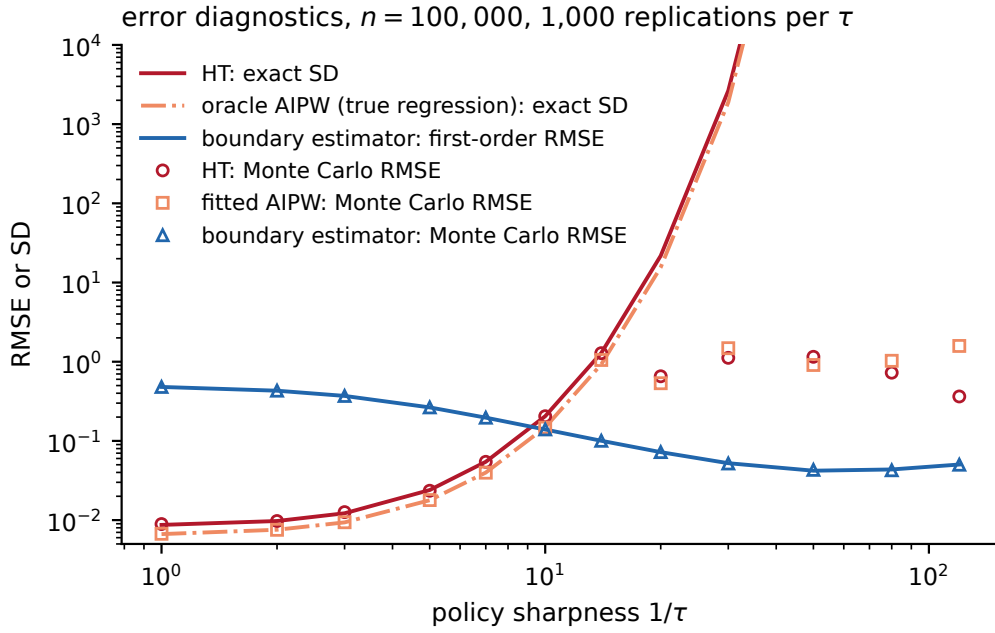


Figure 6: Error diagnostics for the runs of Figure 4. Lines: exact standard deviation of the Horvitz–Thompson estimator and of an oracle augmented estimator that uses the true regression (a reference, not an estimator used in the simulation), both for θ , and the first-order approximation to the root mean squared error of the boundary estimator of β_0 . Markers: Monte Carlo root mean squared error of the Horvitz–Thompson estimator, the augmented estimator with a fitted linear working model, and the boundary estimator.